Capstone Project: AI-Assisted Executive Deck Generation

NovaRetail Quarterly Business Review

Problem Statement

NovaRetail, a multi-region consumer electronics retailer, has just completed its quarterly business review. Leadership needs a concise, 5-slide executive deck covering performance, customer experience, operational risk, and recommended actions, built from raw data rather than handed to you pre-written.

You are given a fixed 5-slide PowerPoint template with named placeholder shapes, a synthetic business dataset, and a small icon library. Your job is to analyze the data, decide what each slide should say and show, and programmatically assemble the final deck, the way a data/BI engineer would automate a recurring reporting workflow.

What You're Given

File	Purpose
novaretail_template.pptx	5-slide template. Every fillable region is a named shape (e.g. title_2, chart_performance, insights_risk, icon_3). Dashed boxes mark where a chart or icon goes; italic bracketed text marks where written content goes.
novaretail_dataset.xlsx	3 sheets, Regional_Performance, Customer_Experience, Operational_Risk, each with 4 regions × 4 quarters of data.
novaretail_icon_repo.zip	8 icons (growth, customer, risk, action, logistics, region, returns, security). Pick the one that best fits each slide's theme, don't just use them in order.

What's Expected From You

- Explore the dataset and decide, per slide, what the real story is (which region is growing, which is at risk, etc.), don't just describe the numbers, interpret them.
- Find each named placeholder in the template programmatically and replace it:
  - Write an AI-generated (or your own) insight-led headline, summary, and bullet points into the text placeholders.
  - Build an appropriate chart (you choose the chart type and what it shows) and insert it at the chart placeholder's exact position/size, pulling data from the provided dataset.
  - Select the one icon from the repo that best matches each slide's theme and insert it at the icon placeholder.
  - Fill the KPI callouts on Slide 1 with values you compute from the dataset.
- Do this with Python and python-pptx, not by hand in PowerPoint. You are expected to explore the python-pptx library yourself (shapes, shape.name, text frames, native charts via GraphicFrame, pictures, positioning) rather than being given a line-by-line script. Reading its documentation/source is part of the assignment.
- Remove each placeholder's dashed box/instruction text once you've replaced it, so the final deck looks like a real executive presentation, not a worked-on template.

Submission

Push your working project (code + the generated/updated .pptx) to a GitHub repository. The repo must contain the final, generated PowerPoint file, not just your script.

Include a short README.md explaining how to run your code and regenerate the deck.

Submit the link to your GitHub repo.

A Few Ground Rules

1. The 5 slides and their layout are fixed, don't add, remove, or resize slides.
2. Every shape name in the template is intentional; use it to locate that shape rather than guessing coordinates.
3. Chart type, headline wording, icon choice, and exact KPI framing are your decisions, that judgment is what's being assessed, not just whether the code runs.